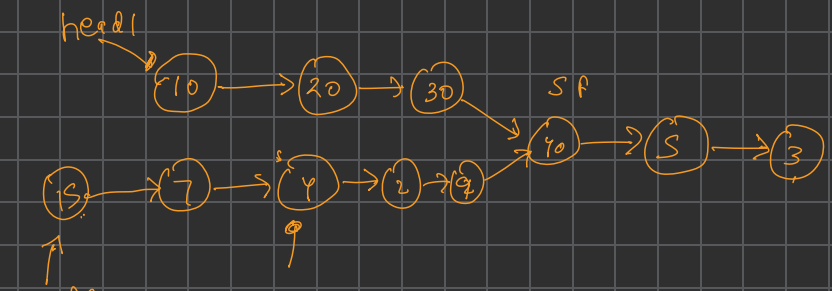
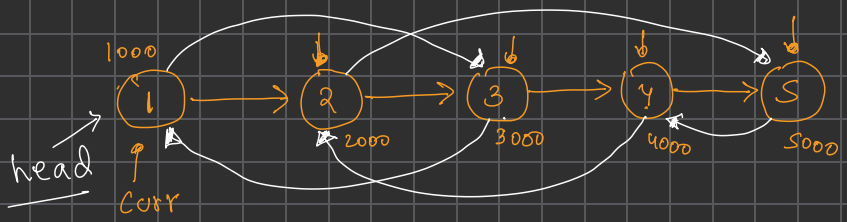
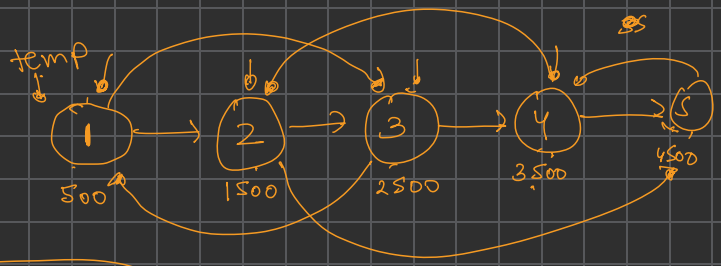


head1 = 6 2 2

head2 = 8

node = data
= next
→ random



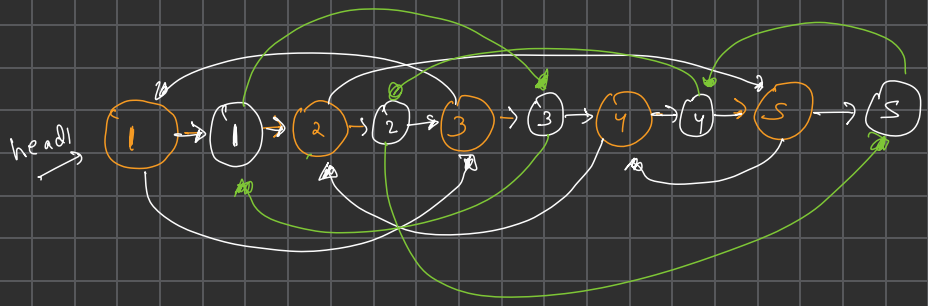


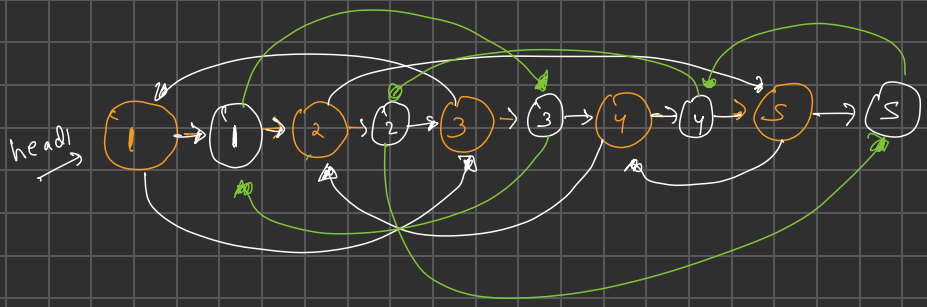
$O(n^2)$

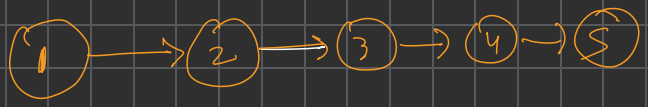
$\boxed{\cancel{O(n)}}$

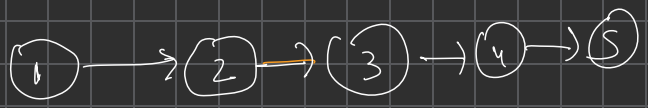
$O(n)$

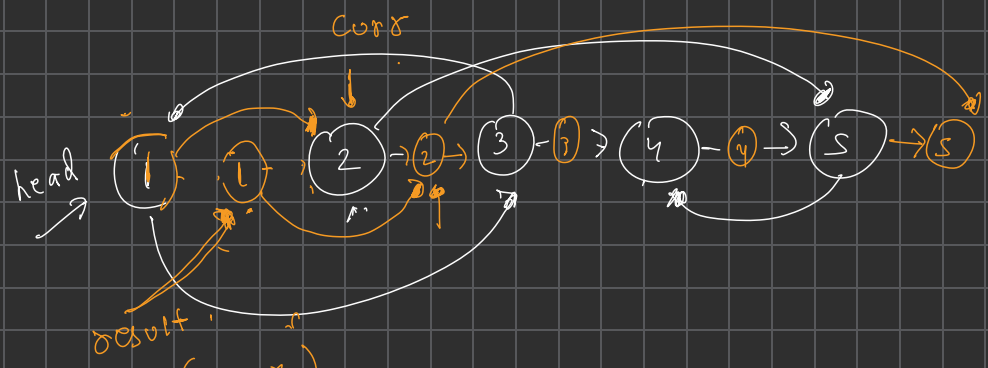
1000	→ 500
2000	→ 1500
3000	→ 2500
4000	→ 3500
5000	→ 4500


head
1
1
2
2
3
3
4
4
5
5


head
1
1
2
2
3
3
4
4
5
5

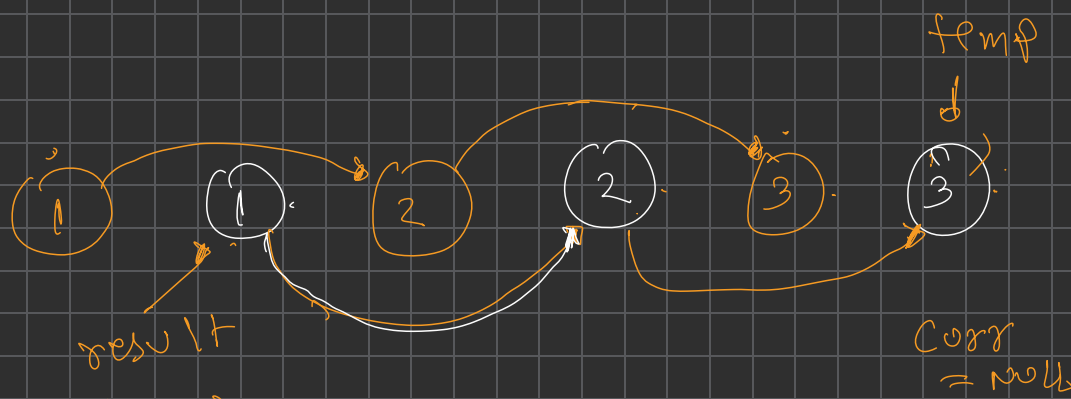

1
2
3
4
5


1
2
3
4
5




```
while (curr)
{
    curr -> next = temp -> next;
    curr = curr -> next;
    temp -> next = curr -> next;
    temp = curr -> next;
}
```

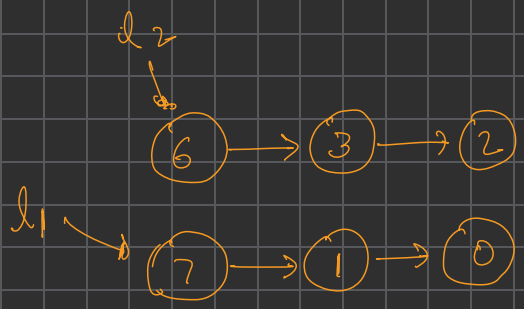
if (curr)




```
while (curr)
{
    curr -> next = temp -> next
    curr = temp -> next;
    if (curr)
    { temp -> next = curr -> next;
      temp = temp -> next; }

}
```

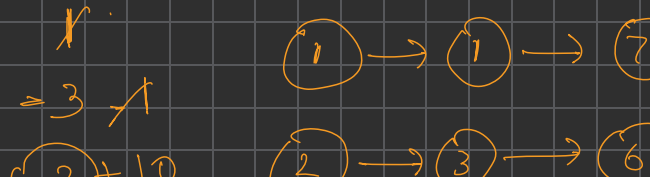
① remove the start zero from both



$$\begin{array}{r} 710 \\ 63 \\ \hline 647 \end{array}$$

7 − 6 = 1

diff = 0



Node * result = NULL;

int borrow = 0

7 − 2 − 0

⇒ −1 + 10

9 9

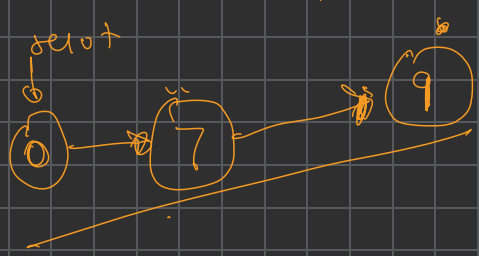
−3 + 10

7

```
while (l1)
{
    int val1 = l1 -> data;
    int val2 = l2 ? l2 -> data : 0;
    int diff = val1 - val2 - borrow;

    if (diff < 0)
    {
        borrow = 1
        diff = diff + 10;
    }
    else
    {
        borrow = 0;
    }

    Node * temp = new Node(diff)
    temp -> next = result;
        result = temp;
```



```
l1 = l1 -> next;
if (l2)
{
    l2 = l2 -> next;
}
}
```

borrow = 1

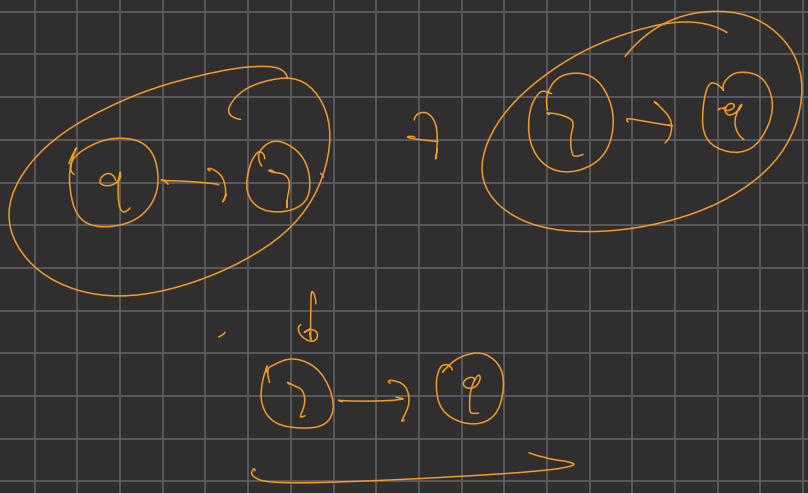
```
  7  2  1
  -  6  8
  -------
  6  5  3
```

1 - 8

-7 + 10

2 - 6 - 1

-5 + 10

7 - 0 - 1

= 6